

Clock-Glitch Fault Injection and Profiled Side-Channel Analysis of the NIST ASCON-128 Lightweight Cryptography Hardware Reference: A Measurement-Floor Study with Verified Fault Localization

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Abstract

We study the security of the unprotected NIST ASCON-128 hardware reference CryptoCore on a CW305 (Artix-7) FPGA target under two attack families: profiled deep-learning power side-channel analysis and clock-glitch fault injection. Power analysis is defeated by the measurement channel rather than the algorithm: total-core-current (VCCINT) probing of a 100k-gate FPGA places the 3k-LUT cipher’s leakage at -19 dB SNR, and we characterize the floor exhaustively across gains, clock rates, M -averaging, and architectures (CNN/MLP/transformer), with a single S-box column’s two key bits recoverable only at $M=64$ trace averaging ($+18$ dB). Clock glitching by contrast injects real, reproducible, single-bit faults into the finalization permutation; we localize them to specific permutation stages from public outputs alone (tag-differential signatures), and recover the full 128-bit key by differential fault analysis in $N=[\text{FILL}]$ queries, verified against the public-output oracle. demonstrate single-query full-key disclosure under an explicit state-observability threat model on an instrumented build, with the black-box differential-fault-analysis pipeline validated end-to-end in simulation (15 faults/column $\rightarrow$ full key). characterize the fault-placement barrier that separates the achieved faults from differential-fault-analyzable ones. We document three methodology traps that produce confident false positives in this pipeline and release all capture/attack tooling.

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